Ningna Wang

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My research interests broadly lie in **computer graphics**, **geometry processing**, and **3D shape analysis**. My current research direction focuses on **3D meshing**, **3D reconstruction** and **3D shape analysis**. Additionally, I am also interested in exploring **aerial path planning** for 3D urban scene reconstruction and **generative 3D shape synthesis**.

Research Positions

Postdoctoral Research Scientist, Columbia University — New York, NY, USA
Advisor: Dr. Silvia Sellán — *2025 - present*

- Developing a series of robust methods to **extract surface mesh** from implicit fields.

PhD Student, The University of Texas at Dallas — Dallas, Texas, USA
Advisor: Dr. Xiaohu Guo — *2019 - 2025*

- Developed a complete framework for computing **high-quality** medial meshs of 3D models with **structure-awareness**.
- Developed a complete framework for computing the medial axis of 3D CAD meshes with **sharp-features preservation**.
- Developed a novel **topology-preserving** 3D medial axis computation framework based on volumetric restricted power diagram (RPD).
- Studied **learning-based methods** for mesh reconstruction via 3D skeleton prediction from **single-view images** or **point clouds**.
- Developed a new method for estimating **globally consistent normal orientations** for a raw point cloud.
- Studied a smooth **mesh simplification** functional that simultaneously consolidates weak features in a high-quality mesh.

Research Intern, Shenzhen University — Shenzhen, China
Advisor: Dr. Hui Huang — *10/2023 - 12/2023*

- Developed an **aerial path planning** method for high-quality digital 3D scene updation.

Degrees

Ph.D. in Computer Science	**University of Texas at Dallas** | Dallas, Texas, USA	*2025*
M.S. in Computer Science	**Carnegie Mellon University** | Pittsburgh, PA, USA	*2016*
B.S. in Computationl Mathematics	**Jilin University** | Changchun, Jilin, China	*2014*

Awards

2025	**EECS Rising Stars**	Massachusetts Institute of Technology (MIT)	USA
2025	**Rising Stars in Computer Graphics**	ACM WiGRAPH	USA
2023	**SIGGRAPH Technical Best Paper Award**	ACM SIGGRAPH	USA
2025	**CVM 2025 Best Paper Award**	Computational Visual Media Conference	HK, China
2024	**Betty and Gifford Johnson Travel Awards**	The University of Texas at Dallas	USA
2024	**First Place CAST STAR Award (1/16 teams)**	CAST-STEM Bridge Summer Camp	USA
2013	**Honorable Mention**	Mathematical Contest in Modeling (MCM)	USA
2010	**First/Second Prize Scholarship & Outstanding Student**	Jilin University (2010-2014)	China

Publications

[Journal & Conference Articles] * denotes equal contributions

[1] MATStruct: High-quality Medial Mesh Computation via Structure-aware Variational Optimization
Ningna Wang, Rui Xu, Yibo Yin, Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo
ACM SIGGRAPH Asia. Hong Kong, China, 2025: pp. 1–12.

[2] MATTopo: Topology-preserving Medial Axis Transform with Restricted Power Diagram **[Journal]**
Ningna Wang, Hui Huang, Shibo Song, Bin Wang, Wenping Wang, Xiaohu Guo
ACM Transactions on Graphics (TOG), 2024, 43(6): pp. 1–16.

[3] Computing Medial Axis Transform with Feature Preservation via Restricted Power Diagram **[Journal]**
Ningna Wang, Bin Wang, Wenping Wang, Xiaohu Guo
ACM Transactions on Graphics (TOG), 2022, 41(6): pp. 1–18.

[4] Aerial Path Online Planning for Urban Scene Updation
Mingfeng Tang*, **Ningna Wang***, Ziyuan Xie, Jianwei Hu, Ke Xie, Xiaohu Guo, Hui Huang
ACM SIGGRAPH North America. Vancouver, Canada, 2025: pp. 1–11.

[5] Dual Contouring of Signed Distance Data
Xiana Carrera, **Ningna Wang**, Christopher Batty, Oded Stein, Silvia Sellán
ACM SIGGRAPH North America. Los Angeles, USA, 2026.

[6] DC-SQEM: Medial Mesh Extraction by Dual Contouring with Spherical Quadric Error Metric **[Journal]**
Bingchuan Li, Ruilong Liu, Tianhao Hu, Congyi Zhang, **Ningna Wang**, Junfeng Yao, Xiaohu Guo
to appear in ACM Transactions on Graphics (TOG), 2026.

[7] Globally Consistent Normal Orientation for Point Clouds by Regularizing the Winding-Number Field **[Best Paper Award]**
Rui Xu, Zhiyang Dou, **Ningna Wang**, Shiqing Xin, Shuangmin Chen, Mingyan Jiang, Xiaohu Guo, Wenping Wang, Changhe Tu
ACM Transactions on Graphics (TOG), 2023, 42(4): pp. 1–15.

[8] CWF: Consolidating Weak Features in High-quality Mesh Simplification **[Journal]**
Rui Xu*, Longdu Liu*, **Ningna Wang**, Shuangmin Chen Chen, Shiqing Xin, Xiaohu Guo, Zichun Zhong, Taku Komura, Wenping Wang, Changhe Tu
ACM Transactions on Graphics (TOG), 2024, 43(4): pp. 1–14.

[9] MVFusion-GS: Motion-Variance Guided Temporal Attention for High-Quality Dynamic Gaussian Splatting
Jianwei Hu, Tingxuan Huang, Hengyu Zhou, **Ningna Wang**, Xiaohu Guo Jinshan Lai, Bin Wang
European Conference on Computer Vision (ECCV), 2026.

[10] NASM: Neural Anisotropic Surface Meshing
Hongbo Li, Haikuan Zhu, Sikai Zhong, **Ningna Wang**, Cheng Lin, Xiaohu Guo, Shiqing Xin, Wenping Wang, Jing Hua, Zichun Zhong
ACM SIGGRAPH Asia. Tokyo, Japan, 2024: pp. 1–12.

[11] CADDreamer: CAD Object Generation from Single-view Images
Yuan Li, Cheng Lin, Yuan Liu, Xiaoxiao Long, Chenxu Zhang, **Ningna Wang**, Xin Li, Wenping Wang, Xiaohu Guo
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). Nashville, USA, 2025.

[12] TexHOI: Reconstructing Textures of 3D Unknown Objects in Monocular Hand-Object Interaction Scenes **[Journal]**
Alakh Aggarwal, **Ningna Wang**, Xiaohu Guo
IEEE Transactions on Visualization & Computer Graphics (TVCG), 2025, 31(10): pp. 8347–8358.

[13] S3DS: Self-supervised Learning of 3D Skeletons from Single View Images
Jianwei Hu, **Ningna Wang**, Baorong Yang, Gang Chen, Xiaohu Guo, Bin Wang
Proceedings of the 31st ACM International Conference on Multimedia (ACM MM). Ottawa, Canada, 2023: pp. 6948–6958.

[14] RBF-MAT: Computing medial axis transform from point clouds by optimizing radial basis functions **[Journal]**
Mengyuan Ge, Junfeng Yao, Baorong Yang, **Ningna Wang**, Zhonggui Chen, Xiaohu Guo
Computer Aided Geometric Design (CAGD), 2025, 118(C): p. 102421.

[15] Point2MM: Learning medial mesh from point clouds **[Journal]**
Mengyuan Ge, Junfeng Yao, Baorong Yang, **Ningna Wang**, Zhonggui Chen, Xiaohu Guo
Computers & Graphics (C&G), 2023, 115(C): pp. 511–521.

[16] IMMAT: Mesh Reconstruction from Single View Images by Medial Axis Transform Prediction **[Journal]**
Jianwei Hu, Gang Chen, Baorong Yang, **Ningna Wang**, Xiaohu Guo, Bin Wang
Computer-Aided Design (CAD), 2022, 150: p. 103304.

Invited Talks

Contouring and Operations on Implicit Geometry
nTop | New York, NY, USA *06/2026*

Inside Shapes: Understanding Shapes through Medial Geometry
University of Toronto | Toronto, Canada *04/2026*
New England Symposium on Graphic | MIT, Cambridge, MA, USA *04/2026*
GAMES Webinar | Online *04/2026*

Computing 3D Medial Axis Transform via Restricted Power Diagram (RPD)
Visual Computing Research Center, Shenzhen University | Shenzhen, China *12/2025*

MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization
ACM SIGGRAPH Asia 2025 | HongKong, China *12/2025*
Center for Digital Media Computing, Xiamen University | Online *12/2025*

Aerial Path Online Planning for Urban Scene Updation
ACM SIGGRAPH 2025 | Vancouver, Canada *08/2025*

MATTopo: Topology-preserving Medial Axis Transform with Restricted Power Diagram
ACM SIGGRAPH ASIA 2024 | Tokyo, Japan *12/2024*
Visual Computing Research Center, Shenzhen University | Shenzhen, China *11/2023*
Center for Digital Media Computing, Xiamen University | Online *10/2024*

Computing Medial Axis Transform with Feature Preservation via Restricted Power Diagram
ACM SIGGRAPH ASIA 2022 | Daegu, South Korea *12/2022*
Center for Digital Media Computing, Xiamen University | Online *01/2024, 11/2022*

Other Work Experience

Booking.com B.V. Amsterdam, Netherlands
Senior Software Engineer *2018 - 2019*
Software Engineer *2017 - 2018*

- Designed, built, and operated the **hotel availability search system**, serving thousands of search requests per second.
- Improved system stability and scalability by distributing hotel availabilities with **jump consistent hashing**.
- Collaborated with product and partner engineers to develop and deliver the search system.

The Priceline Group Inc. Amsterdam, Netherlands | Seattle, WA, USA
Graduate Software Engineer *2016 - 2017*

- Developed a **hotel inventory management system** for property listing, yield management, and revenue analytics.
- Implemented and tested new features for the **Genius loyalty program**, including discounts and travel rewards.

Teaching

Mentor	MIT Summer Geometry Initiative	*2026 Summer*
Teaching Assistant	UTD CS4361/CS6366 Computer Graphics	*2025 Spring, 2024 Fall, 2021 Spring*
Teaching Assistant	UTD Clark Summer Research Program	*2024 Summer*
Teaching Assistant	CAST-STEM Bridge Summer Camp	*2024 Summer*

- Supervisor of a 9-high-school-students team on a 6-weeks 3D talking face project.
- Instructor of 2-weeks full-day deep learning tutorials including lectures, coding demos and exercises.

Teaching Assistant	UTD CS6323 Computer Animation and Gaming	*2022 Fall*
Teaching Assistant	UTD CS4347 Database Systems	*2022 Fall, 2021 Spring*
Teaching Assistant	UTD CS6334 Virtual Reality	*2020 Spring*
Teaching Assistant	UTD CS4332 Introduction to Programming Video Games	*2019 Fall*

Academic Service

Conference	Program Committee	**ACM SIGGRAPH North America**	*2027*
Conference	Posters Jury	**ACM SIGGRAPH North America**	*2026*
Conference	Program Committee	**Pacific Graphics**	*2026*
Conference	Program Committee	**Symposium on Solid and Physical Modeling (SPM)**	*2026*
Conference	Program Committee	**Computer Graphics International (CGI)**	*2025*
Conference	Program Committee	**International Conference on Computational Visual Media (CVM)**	*2024*
Conference	Reviewer	**ACM SIGGRAPH North America**	*2026, 2025, 2024*
Conference	Reviewer	**ACM SIGGRAPH Asia**	*2026, 2025, 2024*
Conference	Reviewer	**Pacific Graphics**	*2026, 2023*
Conference	Reviewer	**Eurographics**	*2026*
Conference	Reviewer	**International Conference on 3D Vision (3DV)**	*2026*

Conference	Reviewer	**International Conference on Geometric Modeling and Processing (GMP)**	*2024*
Journal	Reviewer	**IEEE Transactions on Visualization and Computer Graphics (TVCG)**	*2026, 2025, 2022*
Journal	Reviewer	**Computer Aided Geometric Design (CAGD)**	*2024*
Journal	Reviewer	**Graphical Models (GMOD)**	*2025, 2024*
Journal	Reviewer	**IEEE Computer Graphics and Applications**	*2026*

Research Mentoring

I co-mentored these students alongside their own faculty advisors on research projects (theirs or mine); all papers are co-authored with me.

Xiana Carrera	PhD student, Columbia University · ***Papers: SIGGRAPH 2026; 2 under review***	*2025–present*
Beatrice Stotz	Undergraduate student, Columbia University · ***Papers: 1 short paper under review***	*2025–present*
Eva Kato	Visiting student, Columbia University; now Braid Technologies · ***Papers: SCF 2026 (poster)***	*2025*
Zilong Wang	PhD student, The University of Texas at Dallas · ***Papers: 1 under review***	*2025–present*
Bingchuan Li	PhD student, Xiamen University · ***Papers: TOG 2026***	*2025–present*
Ruilong Liu	MS 2025, now PhD student, Xiamen University · ***Papers: TOG 2026***	*2025–present*
Tianhao Hu	MS student, Xiamen University · ***Papers: TOG 2026***	*2025–2026*
Mingfeng Tang	MS student, Shenzhen University · ***Papers: SIGGRAPH 2025***	*2024–2026*
Ziyuan Xie	MS student, Shenzhen University · ***Papers: SIGGRAPH 2025***	*2024–2025*
Yibo Yin	BS 2025, Wuhan University; now MS student, EPFL · ***Papers: SIGGRAPH Asia 2025***	*2024–2025, 2026–present*
Mengyuan Ge	PhD student, Xiamen University · ***Papers: C&G 2023, CAGD 2025***	*2022–2024*
Shibo Song	MS 2023, Xiamen University; then miHoYo · ***Papers: ACM TOG 2024***	*2021–2024*

Software

Additionally, open-source code is available for publications above at https://ningnawang.github.io/.

LibMAT github.com/ningnawang/libmat

A C++ library for processing the 3D medial axis transform (MAT), with an emphasis on medial meshes.

Blender-mat-addon github.com/songshibo/blender-mat-addon

A Blender plugin for importing and exporting medial meshes (MA format) and interpolating spheres, cones, and slabs.

Q-MAT open-source github.com/ningnawang/QMAT

Open-source C++ code for the SIGGRAPH 2015 paper *Q-MAT: Computing Medial Axis Transform by Quadratic Error Minimization*.

References available upon request.